

# Predictive Model Plan

---

## 1. Model Logic (Generated with GenAI)

The goal of this predictive model is to identify customers who are most likely to become delinquent on their credit payments. The model uses customer financial details, credit history, and payment behavior to estimate the probability of delinquency. By analyzing patterns in historical customer data, the model learns the relationship between different financial factors and the likelihood of missed or late payments.

### Model Workflow

1. Collect customer data such as age, income, credit score, credit utilization, loan balance, debt-to-income ratio, employment status, account tenure, and monthly payment history.
2. Clean the dataset by handling missing values, removing duplicate records, and correcting inconsistent category names.
3. Convert categorical variables into numerical values so they can be used by the machine learning algorithm.
4. Split the dataset into training and testing sets.
5. Train the prediction model using the training data.
6. Test the model using unseen customer records.
7. Generate a probability score that indicates the customer's risk of becoming delinquent.
8. Classify each customer as **Low Risk**, **Medium Risk**, or **High Risk**.

---

Paste your GenAI-generated output below or describe the logic in your own word

---

The predictive model is designed to identify customers who are most likely to become delinquent by analyzing their financial and credit-related information. The model uses customer attributes such as **Income**, **Credit Score**, **Credit Utilization**, **Missed Payments**, **Loan Balance**, **Debt-to-Income Ratio**, **Employment Status**, **Account Tenure**, and **Monthly Payment History** as input features. Before model training, the dataset is preprocessed by handling missing values, removing duplicate records, and converting categorical variables into numerical format.

After preprocessing, the dataset is divided into training and testing sets. A **Random Forest Classifier** is trained using the historical customer data because it can capture complex relationships between multiple financial variables and provides strong prediction accuracy. Once the model is trained, it predicts the probability that each customer will become delinquent. Based on this probability, customers are classified into **Low Risk**, **Medium Risk**, or **High Risk** categories.

### Model Logic

---

1. Load the customer credit dataset.
2. Handle missing values and remove duplicate records.
3. Convert categorical variables into numerical values.
4. Select important features such as Credit Score, Income, Credit Utilization, Missed Payments, Loan Balance, Debt-to-Income Ratio, Employment Status, Account Tenure, and Monthly Payment History.
5. Split the dataset into training (80%) and testing (20%) sets.
6. Train the Random Forest Classifier using the training data.
7. Test the model on unseen customer records.
8. Predict the probability of customer delinquency.
9. Classify customers into **Low Risk**, **Medium Risk**, or **High Risk** based on the prediction score.
10. Evaluate the model using **Accuracy, Precision, Recall, F1 Score, ROC-AUC, and Confusion Matrix**.

This model helps financial institutions identify high-risk customers early, allowing them to take preventive actions, improve credit decisions, and reduce potential financial losses.

---

## 2. Justification for Model Choice

For this project, I selected the **Random Forest Classifier** as the main prediction model.

Random Forest is suitable because customer delinquency is influenced by many financial and behavioral factors that interact with each other. Unlike simple linear models, Random Forest can capture these complex relationships and usually provides higher prediction accuracy. It is also less likely to overfit because it combines the predictions of many decision trees instead of relying on a single tree.

Another advantage is that Random Forest provides **feature importance**, allowing us to identify which customer characteristics have the greatest influence on delinquency prediction. This helps both technical teams and business managers better understand the model's decisions.

Although Logistic Regression is easier to explain, Random Forest is more appropriate for this dataset because it offers better predictive performance while remaining reliable and relatively easy to monitor.

---

## 3. Evaluation Strategy

After training the model, its performance will be evaluated using several standard machine learning metrics.

**Accuracy** will measure the overall percentage of correctly classified customers.

**Precision** will indicate how many customers predicted as delinquent actually became delinquent.

**Recall** will measure how effectively the model identifies customers who are truly at risk. Since missing a high-risk customer can lead to financial loss, recall will be one of the most important evaluation metrics.

**F1 Score** will provide a balance between precision and recall, making it useful if the dataset contains more non-delinquent customers than delinquent customers.

**ROC-AUC Score** will evaluate how well the model separates delinquent customers from non-delinquent customers across different decision thresholds.

A **Confusion Matrix** will also be used to understand the number of correct and incorrect predictions, including false positives and false negatives.

### **Bias Detection and Ethical Considerations**

The model should be checked regularly to ensure that it performs fairly across different customer groups, such as employment status or location. Sensitive personal information should not be used in a way that creates unfair decisions. The model should support human decision-making rather than replace it completely, and all predictions should be reviewed carefully before making important financial decisions.

---

## **Recommended Input Features**

The five most important features for predicting customer delinquency are:

1. Credit Score
2. Credit Utilization
3. Missed Payments
4. Debt-to-Income Ratio
5. Loan Balance

These variables provide the strongest indication of a customer's financial health and repayment behavior.

---

## **Overall Summary**

This predictive model uses historical customer financial data to estimate the likelihood of future credit delinquency. After comparing different machine learning approaches, **Random Forest**

was selected because it provides better prediction accuracy and can identify complex relationships between customer characteristics. The model will be evaluated using Accuracy, Precision, Recall, F1 Score, ROC-AUC, and a Confusion Matrix to ensure reliable performance. Regular monitoring and fairness checks will also be carried out so that the model remains accurate, transparent, and suitable for practical use in Geldium's credit risk assessment process.